

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Electrical and Electronic Engineering
Summer – 2024 Final Examination
Program: B.Sc. in EEE (Batch – 17th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 0533-1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 10

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

CLO1	Specify the basic terms related to thermodynamics, modern physics and light.	4
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Cognitive Knowledge

1. What is the radius ($\text{\AA}$) of the first orbit (ground state) of a hydrogen atom?
- A) 5.3
 - B) 2.7
 - C) 1.2
 - D) 0.53
 - E) 0.24

Cognitive Understanding

2. The mean life of a radionuclide is _____ the half-life.
- A) twice of
 - B) the reciprocal of
 - C) equal to
 - D) independent of
 - E) approximately 1.44 times of

Cognitive Knowledge

3. What is the equivalent of 1eV in Joules?
- A) 1.6×10^{-19}
 - B) 1.67×10^{-27}
 - C) 1.67×10^{-31}
 - D) 9.1×10^{-27}
 - E) 19.1×10^{-31}

Cognitive Understanding

4. Which is essential for interference?
- A) Two different color lights.
 - B) The movement of the two waves should be opposite.
 - C) The light sources should be far apart.
 - D) The light sources should be coherent.
 - E) The phase difference should have to be $\frac{\pi}{2}$.

CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	3
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Cognitive Understanding

5. At polarizing angle, the angle between the reflected ray and refracted ray from a surface is -
- A) 0°
 - B) 30°
 - C) 60°
 - D) 90°
 - E) 180°

Cognitive Understanding

6. What is the resultant amplitude of destructive interference of two waves with amplitudes a and b ?
- A) $a+b$
 - B) $a-b$
 - C) 0
 - D) $|a-b|$
 - E) ab

Cognitive Knowledge

7. Which of the following phenomena could not be explained by Newton's corpuscular theory of light?
- A) Reflection of light
 - B) Refraction of light
 - C) Interference of light
 - D) Rectilinear propagation of light
 - E) Sensation of vision

CLO3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and light.	2
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Cognitive Applying

8. What is the energy (eV) of an electron of the 3rd orbit of Hydrogen atom?
- A) 0.89
 - B) 1.51
 - C) 3.70
 - D) 13.6
 - E) 17.6

Cognitive Applying

9. Which one will be the angular momentum of 2nd orbit according to the Bohr postulates?
- A) h
 - B) $h/2$
 - C) $h/4$
 - D) $h/2\pi$
 - E) h/π

CLO4	Demonstrate analytical proficiency to establish the physical and mathematical logic behind natural phenomena observed in everyday life.	1
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Cognitive Analyzing

10. Under what condition will the Compton wavelength of a photon be doubled?
- A) When the scattering angle is 90°
 - B) When the scattering angle is 180°
 - C) When the photon's energy is halved
 - D) When the electron is at rest
 - E) When the photon's frequency is doubled

Signature of Examiner

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PART-B: SAQ

Time: 1 Hour and 50 Minutes

Total Marks: 30

[Instruction: Marks for the questions are indicated against each of them]

CLO1	Specify the basic terms related to thermodynamics, modern physics and light.	5
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1. a) Illustrate spectral series of hydrogen atom. [2]
b) Clarify: Electron Capture, Stopping Potential & Half-life. [3]

CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	10
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2. a) Formulate the Poynting vector using proper notation. [2]
b) Explain the formation of Newton's rings. [3]
3. a) Illustrate the energy-frequency curve for the photoelectric effect. [2]
b) Establish the relation between momentum and wave length. [3]

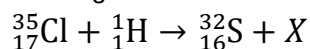
CLO3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and light.	10
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[Answer any 2 (two) out of 3 (three) sets of questions]

4. a) Calculate the energy required to excite the hydrogen atom from the ground state (n=1) to the first excited state (n=2). [2]
b) Show that the velocity of an electron in the 1st orbit in hydrogen atom is $\left(\frac{1}{137}\right)c$, where 'c' is the velocity of light. [3]
5. a) Due to collision of x-ray photon of initial frequency of $3 \times 10^{19} \text{ Hz}$ with an electron, the photon is scattered by 90° . Calculate the new frequency of it. [Compton wavelength = $2.43 \times 10^{-12} \text{ m}$] [2]
b) UV light of wavelength 350 nm and intensity 1.00 W/m^2 is directed at a potassium surface. Find the maximum K.E. of photoelectrons. (Work function of Potassium is 2.2 eV) [3]
6. a) Find the mass of an electron ($m_0 = 9.1 \times 10^{-31} \text{ kg}$) whose velocity is $0.99c$. [2]
b) 1 gm of radium is reduced by 2.1 mg in 5 years by alpha decay. Calculate the half-life period of radium. [3]

CLO4	Demonstrate analytical proficiency to establish the physical and mathematical logic behind natural phenomena observed in everyday life.	5
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7. See a nuclear reaction given in below and answer the following questions:



[Mass of proton = 1.00728 amu, Mass of neutron = 1.00867 amu, Mass of the X = 4.00276 amu]

- a) Symbolize X. [1]
- b) Find the radius of X. [1]
- c) Investigate the mass defect and binding energy of X. [3]